

TOLEDO SUBURBAN, MI, USA

WMO: 720275

Lat: 41.736N

Lon: 83.655W

Elev: 669

StdP: 14.34

Time Zone: -5.00 (NAE)

Period: 04-19

WBAN: 04872

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.1	7.4	-7.6	3.7	2.1	-0.5	5.5	9.4	21.3	33.7	19.0	28.9	4.4	250	0.458

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.4	90.3	72.8	87.9	71.9	85.2	70.8	76.0	85.8	74.5	83.6	73.0	81.5	7.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.9	125.0	81.0	71.9	120.9	80.0	70.3	114.2	78.4	39.9	85.6	38.4	83.8	37.0	82.2	81.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.2	16.1	13.5	DB	-4.3	94.0	5.9	2.9	-8.5	96.1	-12.0	97.8	-15.3	99.4	-19.6	101.5	
			WB	-5.1	78.4	5.6	1.5	-9.1	79.5	-12.4	80.3	-15.5	81.1	-19.6	82.2	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	50.7	26.7	28.0	37.3	48.9	60.4	69.6	72.8	71.0	65.0	53.4	41.8	32.2
	DBStd	18.64	11.38	10.89	10.48	8.79	8.82	6.39	5.79	5.32	7.44	8.79	8.96	9.49
	HDD50	2791	725	620	419	123	11	0	0	0	1	62	276	554
	HDD65	5994	1188	1037	860	487	200	26	6	11	91	375	698	1015
	CDD50	3052	2	3	25	90	335	589	706	651	452	168	28	3
	CDD65	780	0	0	1	4	59	165	247	197	92	15	0	0
	CDH74	7005	0	0	11	72	595	1454	2301	1666	774	128	4	0
	CDH80	2153	0	0	1	7	170	476	782	460	229	28	0	0
Wind	WSAvg	5.5	7.1	6.8	6.5	7.1	5.4	4.7	3.9	3.4	3.6	5.1	6.0	6.6
Precipitation	PrecAvg	33.2	2.1	2.0	2.3	3.1	3.5	3.5	3.3	3.2	2.8	2.5	2.4	2.3
	PrecMax	44.4	3.9	4.9	5.0	6.2	6.3	5.8	7.0	8.3	5.3	6.4	6.8	5.4
	PrecMin	26.0	0.6	0.5	0.6	0.9	1.3	1.1	1.2	0.7	1.1	0.2	0.9	0.5
	PrecStd	4.7	0.9	1.1	1.1	1.2	1.7	1.3	1.4	1.7	1.3	1.4	1.2	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	58.8	63.2	74.7	81.2	88.9	92.8	94.8	90.7	90.8	84.2	71.8	60.9
		MCWB	54.7	54.6	62.3	61.1	70.1	72.6	75.0	73.5	73.8	69.4	57.8	56.6
	2%	DB	52.8	54.6	65.5	75.1	84.7	89.7	90.4	87.9	86.5	77.4	64.1	54.8
		MCWB	49.8	50.4	55.8	58.8	67.2	71.5	73.4	72.5	71.5	64.4	55.2	51.6
	5%	DB	47.5	47.5	59.5	70.5	81.2	85.7	87.8	85.0	81.8	72.7	60.7	49.8
		MCWB	44.2	43.1	51.9	56.7	65.9	69.9	72.7	71.9	68.4	62.1	52.5	46.5
	10%	DB	42.1	41.6	54.2	64.4	76.8	81.8	84.2	82.1	78.7	68.5	55.2	44.8
		MCWB	38.8	37.5	48.3	53.5	64.1	68.0	71.0	69.9	67.4	60.3	49.0	41.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.1	57.0	63.2	64.5	72.6	75.9	78.1	77.6	75.8	72.1	61.7	57.6
		MCDB	58.0	60.9	74.5	76.0	83.9	88.0	89.1	86.3	86.9	81.0	68.0	60.9
	2%	WB	50.7	50.6	58.0	61.2	70.3	73.9	76.4	75.4	73.5	68.0	57.3	52.2
		MCDB	52.4	54.5	63.6	70.9	81.1	84.5	86.5	83.5	82.8	75.0	62.9	54.6
	5%	WB	44.4	42.9	53.3	58.3	68.1	72.4	74.8	73.7	71.5	64.4	53.4	46.2
		MCDB	46.5	47.0	58.5	67.7	77.7	81.9	83.8	81.6	78.7	70.0	58.6	48.9
	10%	WB	38.8	38.0	48.1	55.4	66.0	70.7	73.1	72.3	69.4	61.1	49.9	41.3
		MCDB	41.9	41.6	54.4	64.8	74.8	79.0	81.3	79.6	75.4	67.7	55.2	44.2
Mean Daily Temperature Range	5% DB	MDBR	13.3	14.0	16.0	19.2	19.9	19.3	19.4	19.6	20.6	19.0	15.2	11.5
		MCDBR	17.9	20.7	24.6	27.6	24.6	22.8	21.4	21.9	24.1	24.5	22.2	17.4
		MCWBR	15.9	17.2	16.8	15.3	11.8	10.0	8.9	10.0	11.5	13.7	15.2	15.0
	5% WB	MCDBR	17.2	20.2	22.6	24.4	21.6	19.4	18.0	17.8	19.9	20.5	20.2	16.8
		MCWBR	15.9	17.4	17.1	16.1	11.5	10.1	8.7	9.2	10.5	13.2	15.6	15.4
Clear-Sky Solar Irradiance	taub	0.297	0.334	0.360	0.405	0.432	0.451	0.443	0.433	0.385	0.360	0.322	0.311	
	taud	2.413	2.296	2.313	2.211	2.186	2.179	2.219	2.233	2.346	2.400	2.463	2.399	
	Ebn at Noon	271	274	280	274	269	263	264	263	269	262	259	253	
	Edh at Noon	25	33	36	43	46	46	44	42	35	30	24	23	
All-Sky Solar Radiation	RadAvg	488	776	1110	1467	1716	1918	1927	1685	1370	903	574	398	
	RadStd	50	102	102	126	149	128	105	64	113	75	77	39	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (66 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+157	N/A

Nomenclature: See separate page